



MOCK TEST (WHOLE XII)

➡ 180 MCQs

➡ PMDC pattern

Biology

- Q.1** In mammals, the nephron plays a central role in osmoregulation; identify the specific part where selective reabsorption of glucose, amino acids, and most ions occurs.
- A. Bowman's Capsule
 - B. Loop of Henle
 - C. Proximal Convolted Tubule
 - D. Collecting Duct
- Q.2** During dehydration, which hormone is released to increase water reabsorption and where does it act primarily?
- A. Aldosterone - PCT
 - B. ADH —Collecting Duct
 - C. ADH— Glomerulus
 - D. Cortisol - DCT
- Q.3** Which of the following best explains the role of counter-current mechanism in Loop of Henle?
- A. Facilitates active transport of glucose
 - B. Creates osmotic gradient in medulla for water reabsorption
 - C. Increases GFR through hydrostatic pressure
 - D. Maintains blood glucose homeostasis
- Q.4** The malfunctioning of which renal structure leads to glycosuria in diabetic patients?
- A. Loop of Henle
 - B. Glomerulus
 - C. Proximal Convolted Tubule
 - D. Distal Convolted Tubule
- Q.5** Which homeostatic feedback system maintains blood calcium levels by interaction of PTH and calcitonin?
- A. Positive feedback
 - B. Negative feedback
 - C. Feed forward
 - D. Autoregulation
- Q.6** Which organ is primarily involved in excretion of nitrogenous waste in birds and reptiles, producing uric acid to conserve water?
- A. Kidney
 - B. Skin
 - C. Liver
 - D. Lungs
- Q.7** In marine bony fishes, osmoregulation involves:
- A. Excreting large volume of dilute urine
 - B. Drinking seawater and excreting excess salts via gills
 - C. Storing ammonia in tissues
 - D. Converting ammonia into urea for water conservation
- Q.8** What is the main function of hepatic portal system in homeostasis?
- A. Carrying oxygenated blood to liver
 - B. Transporting absorbed nutrients from intestine to liver
 - C. Regulating calcium level in bones
 - D. Maintaining osmotic balance in nephron
- Q.9** A person suffering from Addison's disease is unable to regulate sodium and potassium balance. Which hormone is deficient?
- A. Insulin
 - B. Aldosterone

- C. ADH
D. Cortisol
- Q.10 Which kidney disorder is directly related to immune response against streptococcal infection causing inflammation of glomeruli?**
A. Uremia
B. Glomerulonephritis
C. Kidney stone
D. Nephrosis
- Q.11 Which component of bone matrix provides rigidity, and which provides flexibility?**
A. Collagen - Rigidity, Calcium phosphate - Flexibility
B. Calcium phosphate - Rigidity, Collagen - Flexibility
C. Osteocytes - Rigidity, Blood vessels - Flexibility
D. Marrow— Rigidity, Periosteum - Flexibility
- Q.12 The functional unit of skeletal muscle responsible for contraction is:**
A. Myofibril
B. Sarcolemma
C. Sarcomere
D. T-tubule
- Q.13 During muscle contraction, calcium binds with:**
A. Myosin
B. Actin
C. Troponin
D. Tropomyosin
- Q.14 Which of the following joints allows movement in all directions such as hip joint?**
A. Hinge joint
B. Ball and socket joint
C. Pivot joint
D. Saddle joint
- Q.15 The mineral that directly regulates muscle contraction by binding with proteins and is stored in sarcoplasmic reticulum is:**
A. Sodium
B. Potassium
C. Calcium
D. Magnesium
- Q.16 Which disorder is caused by Vitamin D deficiency leading to soft and weak bones in children?**
A. Rickets
B. Osteoporosis
C. Arthritis
D. Gout
- Q.17 Smooth muscles differ from skeletal muscles in that:**
A. They are striated and multinucleated
B. They are voluntary and fatigue quickly
C. They are non-striated and involuntary
D. They contract only in presence of oxygen
- Q.18 The major difference between endoskeleton and exoskeleton in animals is:**
A. Endoskeleton is molted while exoskeleton is permanent
B. Exoskeleton is external and protective, endoskeleton internal and supportive
C. Endoskeleton prevents growth while exoskeleton allows growth
D. Both are rigid and external
- Q.19 Which connective tissue connects bone to bone at joints?**
A. Tendon
B. Ligament
C. Cartilage
D. Fascia
- Q.20 Arthritis is characterized by:**
A. Inflammation of joints reducing mobility
B. Mineral deposition in muscles
C. Enlargement of tendons
D. Loss of bone marrow cells
- Q.21 Which part of neuron transmits impulse away from the cell body?**
A. Dendrite
B. Soma
C. Axon
D. Synaptic knob

- Q.22 The resting membrane potential of a neuron is maintained by:**
- A. Diffusion of glucose across membrane
 - B. Na⁺/K⁺ pump and selective permeability
 - C. Constant opening of sodium channels
 - D. Neurotransmitter binding
- Q.23 The part of brain responsible for coordination of movement and balance is:**
- A. Cerebrum
 - B. Cerebellum
 - C. Medulla oblongata
 - D. Hypothalamus
- Q.24 Which neurotransmitter deficiency is linked to Parkinson's disease?**
- A. Dopamine
 - B. Serotonin
 - C. Acetylcholine
 - D. GABA
- Q.25 Saltatory conduction in neurons occurs due to:**
- A. Continuous flow of ions in unmyelinated fibers
 - B. Jumping of action potential between nodes of Ranvier
 - C. Passive diffusion across axolemma
 - D. Increased neurotransmitter secretion
- Q.26 Which meninges layer is most closely associated with brain surface?**
- A. Dura mater
 - B. Arachnoid mater
 - C. Pia mater
 - D. Myelin sheath
- Q.27 Reflex arc differs from voluntary action in that:**
- A. It involves spinal cord without cerebral cortex involvement
 - B. It requires conscious brain activity
 - C. It is always a slow response
 - D. It is independent of receptors
- Q.28 The autonomic nervous system regulates:**
- A. Voluntary skeletal muscle movement
 - B. Reflex actions of spinal cord
 - C. Involuntary activities of internal organs
 - D. Transmission of sensory signals
- Q.29 Which ion influx triggers release of neurotransmitters at synaptic cleft?**
- A. Sodium
 - B. Potassium
 - C. Calcium
 - D. Chloride
- Q.30 Alzheimer's disease is associated with accumulation of:**
- A. Uric acid crystals in neurons
 - B. Amyloid plaques and neurofibrillary tangles
 - C. Myelin sheath regeneration
 - D. Excess dopamine production
- Q.31 Spermatogenesis occurs in:**
- A. Epididymis
 - B. Vas deferens
 - C. Seminiferous tubules
 - D. Prostate gland
- Q.32 Which hormone from anterior pituitary initiates spermatogenesis?**
- A. LH
 - B. FSH
 - C. Prolactin
 - D. Testosterone
- Q.33 Fertilization in human occurs usually in:**
- A. Ovary
 - B. Fallopian tube
 - C. Uterus
 - D. Cervix
- Q.34 The structure formed after union of sperm and egg that undergoes cleavage is:**
- A. Zygote
 - B. Blastula
 - C. Morula
 - D. Gastrula
- Q.35 Menstrual cycle is regulated mainly by:**

- A. Estrogen and progesterone
C. All of these
- B. LH and FSH
D. None of these
- Q.36 Which contraceptive method prevents fertilization mechanically?**
A. Oral pills
C. Condom
B. IUD
D. Vasectomy
- Q.37 Which phase of menstrual cycle is characterized by repair and thickening of endometrium under estrogen influence?**
A. Menstrual phase
C. Ovulation
B. Follicular (proliferative) phase
D. Luteal phase
- Q.38 Which event triggers ovulation?**
A. Peak of FSH
C. Drop in progesterone
B. Peak of LH surge
D. Rise in prolactin
- Q.39 The placenta allows exchange of nutrients and gases between mother and fetus but prevents:**
A. Hormone transfer
C. Oxygen diffusion
B. Blood mixing
D. Waste removal
- Q.40 Which assisted reproductive technique involves direct injection of sperm into egg?**
A. IVF
C. AI
B. ICSI
D. Surrogacy
- Q.41 Mendel's law of segregation states:**
A. Alleles separate during gamete formation
C. Traits blend in offspring
B. Alleles remain linked in gametes
D. Genes segregate independently
- Q.42 Which inheritance pattern results in blending of traits, e.g., red x white flowers producing pink?**
A. Complete dominance
C. Incomplete dominance
B. Codominance
D. Multiple alleles
- Q.43 A man with blood group AB marries a woman with blood group O. Which blood groups are possible in children?**
A. A, B
C. A, B, AB
B. AB, O
D. Only O
- Q.44 Which disorder is caused by absence of clotting factor VIII?**
A. Thalassemia
C. Sickle cell anemia
B. Hemophilia
D. Cystic fibrosis
- Q.45 If a heterozygous tall pea plant is crossed with a dwarf, what percentage of offspring will be tall?**
A. 25%
C. 75%
B. 5%
D. 100%
- Q.46 The phenomenon where one gene influences multiple traits is called:**
A. Epistasis
C. Polygenic inheritance
B. Pleiotropy
D. Codominance
- Q.47 In humans, color blindness is:**
A. Autosomal dominant
C. X-linked recessive
B. Autosomal recessive
D. Y-linked dominant
- Q.48 Which chromosomal abnormality leads to Down's syndrome?**
A. Trisomy 21
C. Trisomy 18
B. Monosomy X
D. Trisomy 13
- Q.49 DNA fingerprinting is based on:**
A. Exon sequences
B. Introns and VNTRs

- C. rRNA sequences
D. Mitochondrial DNA
- Q.50 Which nucleic acid base pairing is correct according to Chargaff's rule?**
A. A = C, T=G
B. A=T, C=G
C. A=G, C=T
D. A=U, C=T
- Q.51 The similarity in structure of forelimbs of humans, bats, and whales represents:**
A. Analogous organs
B. Homologous organs
C. Vestigial organs
D. Convergent evolution
- Q.52 Darwin's theory of natural selection emphasizes:**
A. Use and disuse of organs
B. Survival of fittest based on variations
C. Mutation as sole source of evolution
D. Sudden appearance of new species
- Q.53 Which evidence of evolution is provided by fossil records?**
A. Anatomical evidence
B. Paleontological evidence
C. Embryological evidence
D. Biochemical evidence
- Q.54 Industrial melanism in peppered moths is an example of:**
A. Genetic drift
B. Directional selection
C. Stabilizing selection
D. Artificial selection
- Q.55 Which scientist proposed the theory of inheritance of acquired characteristics?**
A. Darwin
B. Lamarck
C. Mendel
D. Wallace
- Q.56 The Hardy-Weinberg equilibrium assumes absence of:**
A. Random mating
B. Gene flow, mutation, selection
C. Large population size
D. All of these
- Q.57 Vestigial structures like appendix in humans indicate:**
A. Sudden mutation
B. Evidence of evolution from ancestors
C. Active use in modern humans
D. Role in digestion
- Q.58 Adaptive radiation is observed in:**
A. Darwin's finches of Galapagos
B. Peppered moths in England
C. Bacteria becoming resistant
D. Artificially selected crops
- Q.59 Which type of evolution results when unrelated species evolve similar adaptations in similar environments?**
A. Divergent evolution
B. Convergent evolution
C. Coevolution
D. Parallel evolution
- Q.60 Founder effect is a type of:**
A. Mutation
B. Natural selection
C. Genetic drift
D. Gene flow
- Q.61 Restriction enzymes cut DNA at specific:**
A. Ribosomes
B. Recognition sequences
C. Promoters
D. Codons
- Q.62 PCR is used for:**
A. Cutting DNA
B. Amplifying DNA
C. Sequencing proteins
D. Protein synthesis
- Q.63 Which plasmid is widely used as vector in genetic engineering?**
A. Ti plasmid
B. pBR322
C. F plasmid
D. R plasmid
- Q.64 Which technique is used for separating DNA fragments according to size?**

- A. Gel electrophoresis
C. Spectrophotometry
- B. Centrifugation
D. Blotting
- Q.65 Insulin production by recombinant DNA technology uses:**
- A. Human fibroblasts
C. Yeast with insulin plasmid
- B. coli with human insulin gene
D. Pancreatic cell culture
- Q.66 Which blotting technique is used for detecting RNA?**
- A. Southern blotting
C. Western blotting
- B. Northern blotting
D. Eastern blotting
- Q.67 Golden rice was genetically engineered to produce:**
- A. Iron
C. Vitamin C
- B. Vitamin A
D. Protein supplement
- Q.68 The term 'transgenic organism' refers to:**
- A. Naturally mutated species
C. Hybrid organism from cross breeding
- B. Organism with foreign gene inserted
D. Clones of parental genotype
- Q.69 Which enzyme is essential for joining DNA fragments in recombinant technology?**
- A. DNA polymerase
C. Restriction endonuclease
- B. DNA ligase
D. RNA polymerase
- Q.70 The process of making identical copies of an organism is called:**
- A. Cloning
C. Hybridization
- B. Transformation
D. Translation
- Q.71 Hormones are secreted into:**
- A. Lymph only
C. Ducts and vessels
- B. Blood only
D. Directly into bloodstream
- Q.72 Which endocrine gland has both exocrine and endocrine functions?**
- A. Thyroid
C. Pancreas
- B. Pituitary
D. Adrenal
- Q.73 Deficiency of iodine leads to:**
- A. Goiter
C. Addison's disease
- B. Diabetes
D. Rickets
- Q.74 Which hormone regulates basal metabolic rates?**
- A. Insulin
C. Cortisol
- B. Thyroxine
D. ADH
- Q.75 Which gland is called "master gland"?**
- A. Adrenal
C. Pituitary
- B. Thyroid
D. Pancreas
- Q.76 Hypersecretion of growth hormone in adults causes:**
- A. Gigantism
C. Cretinism
- B. Acromegaly
D. Dwarfism
- Q.77 Which hormone stimulates release of glucose from liver glycogen?**
- A. Insulin
C. Epinephrine
- B. Glucagon
D. Cortisol
- Q.78 The adrenal medulla secretes:**
- A. Cortisol and aldosterone
C. Insulin and glucagon
- B. Epinephrine and norepinephrine
D. Thyroxine and calcitonin
- Q.79 Diabetes mellitus is caused due to:**
- A. Hyposecretion of insulin
C. Hyposecretion of thyroxine
- B. Hypersecretion of insulin
D. Hypersecretion of cortisol

- Q.80 Which hormone regulates circadian rhythms and is secreted by pineal gland?**
 A. Cortisol
 B. Melatonin
 C. Serotonin
 D. Aldosterone
- Q.81 Which chemical signal acts locally on nearby cells instead of distant targets?**
 A. Hormone
 B. Paracrine signal
 C. Neurotransmitter
 D. Autocrine signal

Chemistry

- Q.82 Which synthetic dye is classified as an azo dye because of the presence of -N-N- linkage in its structure?**
 A. Methylene blue
 B. Congo red
 C. Malachite green
 D. Indigo
- Q.83 The ability of a dye to impart color depends primarily on:**
 A. Presence of chromophores
 B. Presence of auxochromes
 C. Both chromophores and auxochromes
 D. Molecular weight
- Q.84 Indigo dye is extensively used for coloring:**
 A. Silk
 B. Cotton
 C. Wool
 D. Nylon
- Q.85 Which property makes mordant dyes different from direct dyes?**
 A. Require a binding agent (metal salt) to attach with fiber
 B. Directly absorbed by fabric
 C. Derived from natural sources only
 D. Used only in acidic medium
- Q.86 Which dye shows structural isomerism due to position of -OH group on benzene ring?**
 A. Phenolphthalein
 B. Congo red
 C. Alizarin
 D. Methyl orange
- Q.87 Synthetic adhesives like epoxy resins are preferred in industry because:**
 A. They are water-soluble
 B. They possess strong cohesive and adhesive forces
 C. They are biodegradable
 D. They contain only natural polymers
- Q.88 Which adhesive is produced from condensation polymerization of urea and formaldehyde?**
 A. Epoxy resin
 B. Fevicol
 C. Bakelite
 D. Urea-formaldehyde resin
- Q.89 The primary mechanism of adhesive bonding is:**
 A. Electrostatic attraction
 B. Mechanical interlocking and chemical bonding
 C. Only Van der Waals forces
 D. Ionic bond formation
- Q.90 Contact adhesives such as rubber-based glues work effectively because:**
 A. They require long curing times
 B. They instantly form strong bonds after solvent evaporation
 C. They are based on ionic bonding
 D. They cannot stick nonporous surfaces
- Q.91 Which natural adhesive is obtained from casein protein of milk?**
 A. Starch glue
 B. Animal glue
 C. Casein glue
 D. Latex adhesive

- Q.92 The process of joining small monomers to form large macromolecules without elimination of by-products is:**
- A. Condensation polymerization
B. Addition polymerization
C. Co-polymerization
D. Vulcanization
- Q.93 Nylon-6,6 is prepared by the condensation of:**
- A. Ethylene glycol and terephthalic acid
B. Adipic acid and hexamethylenediamine
C. Caprolactam
D. Styrene
- Q.94 The property of elasticity in rubber is enhanced by:**
- A. Cross-linking with sulfur atoms
B. Addition of plasticizers
C. Heating with acids
D. High pressure polymerization
- Q.95 Which polymer is used in the manufacture of non-stick cookware coating?**
- A. Teflon
B. Bakelite
C. Polystyrene
D. Nylon
- Q.96 Which type of polymer shows crystalline and amorphous regions due to arrangement of chains?**
- A. Thermoplastic polymer
B. Thermosetting polymer
C. Elastomer
D. Natural rubber
- Q.97 Carboxylic acids are stronger acids than alcohols due to:**
- A. Inductive effect of carbonyl group stabilizing conjugate base
B. Greater molecular weight
C. Presence of hydroxyl group only
D. Hydrogen bonding in alcohols
- Q.98 Which reagent can be used to distinguish carboxylic acids from phenols?**
- A. Sodium metal
B. Sodium bicarbonate
C. Bromine water
D. FeCl solution
- Q.99 Decarboxylation of sodium benzoate with soda lime gives:**
- A. Benzaldehyde
B. Benzene
C. Benzoic acid
D. Toluene
- Q.100 Acetic acid reacts with PCl₅ to produce:**
- A. Acetyl chloride, HCl, POCl₃
B. Ethyl acetate, POCl₃
C. Acetone and HCl
D. Acetaldehyde and POCl₃
- Q.101 Which of the following is an aromatic carboxylic acid?**
- A. Acetic acid
B. Formic acid
C. Benzoic acid
D. Oxalic acid
- Q.102 The functional group in aldehydes is represented as:**
- A. -CHO
B. -CO-
C. -COOH
D. -C=O
- Q.103 Which reagent is commonly used to distinguish aldehydes from ketones?**
- A. Fehling's solution
B. Tollen's reagent
C. Both A and B
D. NaOH solution
- Q.104 Aldol condensation is possible in carbonyl compounds having:**
- A. At least one α -hydrogen atom
B. No α -hydrogen atom
C. Aromatic structure only
D. Tertiary carbon atom
- Q.105 Formaldehyde polymerizes easily because:**
- A. It is less volatile
B. It has no α -hydrogen
C. It is highly reactive due to small size and electron deficiency
D. It has strong hydrogen bonds
- Q.106 The reaction of aldehydes with 2,4-DNP produces:**

- A. Yellow/orange precipitate
C. Green color solution
- B. Brown gas
D. No visible change
- Q.107 Which alcohol cannot undergo oxidation under normal conditions?**
A. Primary alcohol
B. Secondary alcohol
C. Tertiary alcohol
D. Allyl alcohol
- Q.108 Dehydration of ethanol in the presence of concentrated H_2SO_4 gives:**
A. Acetaldehyde
B. Acetic acid
C. Ethene
D. Ethyne
- Q.109 Phenols are more acidic than alcohols because:**
A. Larger molecular weight
B. Resonance stabilization of phenoxide ion
C. Stronger hydrogen bonding
D. Presence of carbonyl group
- Q.110 Phenol reacts with bromine water to form:**
A. Bromobenzene
B. 2,4,6-tribromophenol (white ppt)
C. Benzyl bromide
D. Phenyl bromide
- Q.111 Which reagent is used in Lucas test to differentiate primary, secondary, and tertiary alcohols?**
A. HNO_3
B. $\text{ZnCl}_2 + \text{HCl}$
C. FeCl_3
D. NaOH
- Q.112 Alkyl halides undergo nucleophilic substitution because:**
A. Halogen is highly electronegative making carbon atom electrophilic
B. Halogen is electropositive
C. Carbon is nucleophilic
D. They are aromatic in nature
- Q.113 Which type of alkyl halide undergoes $\text{S}_\text{N}1$ reaction most readily?**
A. Primary
B. Secondary
C. Tertiary
D. Allyl
- Q.114 The major product formed when chloromethane reacts with NaOH (aq) is:**
A. Methanol
B. Formaldehyde
C. Methane
D. Methyl ether
- Q.115 Reaction of alkyl halide with alcoholic KOH leads to:**
A. Substitution
B. Elimination giving alkene
C. Polymerization
D. Oxidation
- Q.116 Which alkyl halide is used as refrigerant under trade name "Freon"?**
A. CCl_4
B. CHCl_3
C. CF_2Cl_2
D. $\text{C}_2\text{H}_5\text{Cl}$
- Q.117 Benzene undergoes electrophilic substitution reactions rather than addition because:**
A. Addition would destroy aromatic stability
B. Benzene has weak π -bonds
C. Addition requires high temperature
D. Benzene contains saturated carbons
- Q.118 The substitution of $-\text{NO}_2$ group in benzene is directed to:**
A. Ortho and para positions
B. Meta position
C. All positions equally
D. Only ortho position
- Q.119 Which reaction involves replacement of hydrogen by halogen in presence of Lewis acid catalyst?**
A. Friedel-Crafts alkylation
B. Nitration
C. Halogenation
D. Sulfonation
- Q.120 The electrophile in nitration of benzene is:**

- A. NO_2
- B. NO_2^+
- C. NO_3
- D. NO_2 radical

Q.121 The resonance energy of benzene is evidence for:

- A. Presence of single bonds only
- B. Extra stability due to delocalized π -electrons
- C. Instability of cyclic compounds
- D. Aromaticity violation

Q.122 Alkanes are generally unreactive because:

- A. They contain weak bonds
- B. They are saturated and non-polar
- C. They contain π -bonds
- D. They are aromatic

Q.123 Which reaction converts alkane into halogenated derivative by free radical substitution?

- A. Hydrogenation
- B. Halogenation in UV light
- C. Cracking
- D. Isomerization

Q.124 Alkynes differ from alkenes because:

- A. They are less reactive
- B. They undergo substitution easily
- C. They contain sp-hybridized carbon atoms with triple bond
- D. They show resonance

Q.125 The colored compounds of transition metals are due to:

- A. s-orbital electrons
- B. d-d electronic transitions within partially filled d-orbitals
- C. Presence of p-orbitals
- D. Absence of electrons

Q.126 Which property is common to all macromolecules such as proteins, nucleic acids, and polysaccharides?

- A. They are small molecules with low molar mass
- B. They are formed by polymerization of repeating units
- C. They are always soluble in water
- D. They contain only carbon and hydrogen

Physics

Q.127 In an AC circuit, the average power consumed is given by:

- A. VI
- B. $VI\cos\theta$
- C. I^2R
- D. V^2/X

Q.128 The inductive reactance of a coil depends on:

- A. Resistance only
- B. Frequency and inductance
- C. Capacitance only
- D. Voltage applied

Q.129 In a purely capacitive AC circuit, the current:

- A. Lags voltage by 90°
- B. Leads voltage by 90°
- C. Is in phase with voltage
- D. Is independent of frequency

Q.130 The resonant frequency of an LCR series circuit is obtained when:

- A. $X_L = X_C$
- B. $X_L = R$
- C. $X_C = R$
- D. $X_L = X_C + R$

Q.131 The power factor of a circuit is unity when it is:

- A. Purely inductive
- B. Purely capacitive
- C. Purely resistive
- D. At resonance

Q.132 In a transformer, losses due to eddy currents are minimized by:

- A. Laminating the core
- B. Using thick copper wires
- C. Increasing the number of turns
- D. Cooling with water

Q.133 Magnetic flux is measured in:

- A. Weber
- C. Henry

- B. Tesla
- D. Ampere-turn

Q.134 The permeability of free space has units:

- A. N/A^2
- B. T.m/A
- C. H/m
- D. All of these

Q.135 The force on a current-carrying conductor in a magnetic field is maximum when the angle between current and field is:

- A. 0°
- B. 45°
- C. 90°
- D. 180°

Q.136 Which law states that magnetic field around a current-carrying conductor is proportional to current and inversely proportional to distance?

- A. Ampere's law
- B. Coulomb's law
- C. Faraday's law
- D. Biot-Savart law

Q.137 The earth's magnetic field protects life from:

- A. Infrared radiation
- B. Cosmic rays and charged particles
- C. Sound waves
- D. UV light

Q.138 Which material is strongly attracted by magnetic field and retains magnetism permanently?

- A. Paramagnetic
- B. Diamagnetic
- C. Ferromagnetic
- D. Non-magnetic

Q.139 Faraday's law of electromagnetic induction states that:

- A. EMF induced is proportional to resistance
- B. EMF induced is proportional to rate of change of magnetic flux
- C. EMF induced depends only on area of loop
- D. EMF induced is always constant

Q.140 Lenz's law is consistent with:

- A. Law of conservation of mass
- B. Law of conservation of energy
- C. Law of conservation of charge
- D. Law of conservation of momentum

Q.141 The SI unit of self-inductance is:

- A. Tesla
- B. Weber
- C. Henry
- D. Ampere

Q.142 In mutual induction, the induced EMF in secondary coil depends upon:

- A. Current in secondary coil
- B. Rate of change of current in primary coil
- C. Resistance of primary coil
- D. Voltage across secondary coil

Q.143 The back EMF in an electric motor is produced because of:

- A. Lenz's law
- B. Coulomb's law
- C. Biot-Savart law
- D. Ohm's law

Q.144 The energy stored in an inductor of inductance L carrying current I is:

- A. LI
- B. $\frac{1}{2} LI^2$
- C. L^2I
- D. I^2/R

Q.145 According to photoelectric effect, electrons are ejected from a metal surface when:

- A. Frequency of light is less than threshold frequency
- B. Energy of photons is greater than work function
- C. Intensity of light is high enough
- D. Voltage is applied externally

Q.146 Which experiment verified the particle nature of electron waves?

- A. Photoelectric effect
C. Davisson-Germer experiment
- B. Compton effect
D. Rutherford scattering
- Q.147 The energy of a photon of wavelength λ is given by:**
A. hc/λ
B. $h\lambda/c$
C. λ/hc
D. $c/h\lambda$
- Q.148 The de Broglie wavelength of a particle is inversely proportional to:**
A. Its energy
B. Its velocity (momentum)
C. Its mass squared
D. Its charge
- Q.149 Which effect proves that light has particle-like momentum?**
A. Interference
B. Diffraction
C. Compton effect
D. Polarization
- Q.150 In Bohr's model, angular momentum of electron in orbit is:**
A. $nh/2\pi$
B. nh
C. $h/2\pi$
D. n^2h
- Q.151 The Balmer series in hydrogen atom lies in:**
A. Infrared region
B. Ultraviolet region
C. Visible region
D. Microwave region
- Q.152 The spectral lines of hydrogen are explained by:**
A. Rutherford model
B. Bohr model
C. Thomson model
D. Quantum mechanical model only
- Q.153 The Rydberg constant is used to calculate:**
A. Ionization energy
B. Energy levels in hydrogen spectrum
C. Binding energy of nucleus
D. Atomic radius
- Q.154 When electron transitions from higher to lower energy level, it:**
A. Absorbs energy
B. Emits a photon of definite wavelength
C. Emits continuous spectrum
D. Stops radiating
- Q.155 The binding energy per nucleon is maximum for:**
A. Hydrogen
B. Uranium
C. Iron
D. Helium
- Q.156 In nuclear fission, heavy nucleus splits into:**
A. Two lighter nuclei with release of energy
B. One heavier nucleus and one lighter nucleus
C. Three identical nuclei
D. Gamma rays only
- Q.157 The uncontrolled chain reaction in nuclear fission leads to:**
A. Electric current
B. Explosion (atomic bomb)
C. Cooling of core
D. Fusion
- Q.158 Which particle is emitted in β^- decay?**
A. Proton
B. Neutron
C. Electron
D. Positron
- Q.159 Nuclear reactors use moderators like heavy water or graphite because:**
A. They increase speed of neutrons
B. They slow down fast neutrons for effective fission
C. They absorb neutrons completely
D. They supply protons
- Q.160 Fusion reaction occurs naturally in:**
A. Earth's core
B. Sun and stars
C. Nuclear reactors
D. Radioactive isotopes

Q.161 A semiconductor diode allows current to flow:

- A. In both directions equally
- B. Only when reverse biased
- C. Only when forward biased
- D. Only at zero potential

Q.162 In a transistor, the current gain (β) is the ratio of:

- A. I_C/I_E
- B. I_B/I_C
- C. I_C/I_B
- D. OI_E/I_B

English

Q.163 Choose the correct indirect form: He said, "I am preparing for the test."

- A. He said that he is preparing for the test.
- B. He said that he was preparing for the test.
- C. He said that he had prepared for the test.
- D. He said that he has been preparing for the test.

Q.164 Select the sentence with the correct use of the conditional:

- A. If I will see him, I will tell him.
- B. If I see him, I will tell him.
- C. If I saw him, I will tell him.
- D. If I see him, I would tell him.

Q.165 Identify the correctly punctuated sentence:

- A. The teacher said "Honesty is the best policy".
- B. The teacher said, "Honesty is the best policy."
- C. The teacher said "Honesty is the best policy."
- D. The teacher, said "Honesty is the best policy".

Q.166 Choose the correct synonym of 'abundant':

- A. Scarce
- B. Rare
- C. Plentiful
- D. Limited

Q.167 Fill in the blank: She is not interested playing video games.

- A. on
- B. for
- C. in
- D. about

Q.168 Choose the sentence with correct subject-verb agreement:

- A. The list of items are on the table.
- B. The list of items is on the table.
- C. The lists of item is on the table.
- D. The lists of items is on the table.

Q.169 Pick the correctly spelled word:

- A. Accomodate
- B. Accomodate
- C. Accommodate
- D. Acomodate

Q.170 Choose the correct antonym of 'benevolent':

- A. Kind
- B. Generous
- C. Malevolent
- D. Merciful

Q.171 Choose the sentence that is grammatically correct:

- A. Neither of the boys were late for class.
- B. Neither of the boys was late for class.
- C. Neither of the boy was late for class.
- D. Neither boys was late for class.

Logical reasoning

Q.172 If all roses are flowers, and some flowers fade quickly, which of the following must be true?

- A. All roses fade quickly.
- B. Some roses may fade quickly.
- C. No flower fades quickly.
- D. Roses are not flowers.

Q.173 Complete the analogy: Book is to Reading as Fork is to:

- A. Writing
- B. Drawing
- C. Eating
- D. Cooking

Q.174 Which number comes next in the series? 2, 6, 12, 20, 30, ?

- A. 36
- C. 42

- B. 40
- D. 56

Q.175 If in a certain code, MEDICAL is written as NCFJDBM, then PATIENT will be written as:

- A. QBUJFOS
- C. QBUJFOU

- B. QBUJFNU
- D. OZSJDMs

Q.176 A clock shows 3:15. What is the angle between the hour and minute hands?

- A. 7.5°
- C. 22.5°

- B. 15°
- D. 30°

Q.177 Statement: All doctors are educated. Some educated are teachers.

Conclusion I: Some teachers may be doctors.

Conclusion II: All teachers are educated.

Which conclusion(s) follow(s)?

- A. Only I
- C. Both I and II

- B. Only II
- D. Neither I nor II

Q.178 Identify the odd one out:

- A. Square
- C. Triangle

- B. Rectangle
- D. Circle

Q.179 Arrange the words to form a logical sequence: 1. College 2. Admission Test 3. School 4. Job 5. Graduation

- A. 3, 2, 1, 5, 4
- C. 3, 1, 2, 5, 4

- B. 2, 3, 1, 5, 4
- D. 1, 2, 3, 5, 4

Q.180 Five friends are sitting in a row. A is to the left of B but right of C. D is to the right of B but left of E. Who is sitting in the middle?

- A. A
- C. C

- B. B
- D. D

ANSWER KEY

Q.NO	ANS	Q.NO	ANS	Q.NO	ANS	Q.NO	ANS	Q.NO	ANS	Q.NO	ANS
1.		31.		61.		91.		121.		151.	
2.		32.		62.		92.		122.		152.	
3.		33.		63.		93.		123.		153.	
4.		34.		64.		94.		124.		154.	
5.		35.		65.		95.		125.		155.	
6.		36.		66.		96.		126.		156.	
7.		37.		67.		97.		127.		157.	
8.		38.		68.		98.		128.		158.	
9.		39.		69.		99.		129.		159.	
10.		40.		70.		100.		130.		160.	
11.		41.		71.		101.		131.		161.	
12.		42.		72.		102.		132.		162.	
13.		43.		73.		103.		133.		163.	
14.		44.		74.		104.		134.		164.	
15.		45.		75.		105.		135.		165.	
16.		46.		76.		106.		136.		166.	
17.		47.		77.		107.		137.		167.	
18.		48.		78.		108.		138.		168.	
19.		49.		79.		109.		139.		169.	
20.		50.		80.		110.		140.		170.	
21.		51.		81.		111.		141.		171.	
22.		52.		82.		112.		142.		172.	
23.		53.		83.		113.		143.		173.	
24.		54.		84.		114.		144.		174.	
25.		55.		85.		115.		145.		175.	
26.		56.		86.		116.		146.		176.	
27.		57.		87.		117.		147.		177.	
28.		58.		88.		118.		148.		178.	
29.		59.		89.		119.		149.		179.	
30.		60.		90.		120.		150.		180.	